Bell Nozzle Rocket Engine Equation Database

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August 19, 2025

$$C_nH_{2n+2} + \frac{3n+1}{2}O_2 \Longrightarrow (n+1)\,H_2O + nCO_2 + energy \tag{1}$$

Propellant Stoichiometry of Hydrocarbon-Oxygen Propellant [1]

$$\dot{m} = \frac{m}{t} \tag{2}$$

Mass Flow Rate

$$\frac{1}{\gamma - 1} = \sum \frac{mol\%}{\gamma_i - 1} \tag{3}$$

Heat Capacity Ratio of a Multi-Compound Gas [2]

$$\frac{P}{P_s} = \left(1 + \frac{\gamma - 1}{2}Ma^2\right)^{-\frac{\gamma}{\gamma-1}} \tag{4}$$

Isentropic Pressure Relation [3]

$$\frac{T}{T_s} = \left(1 + \frac{\gamma - 1}{2}Ma^2\right)^{-1} \tag{5}$$

Isentropic Temperature Relation [3]

$$\frac{\rho}{\rho_s} = \left(1 + \frac{\gamma - 1}{2}Ma^2\right)^{-\frac{1}{\gamma-1}} \tag{6}$$

Isentropic Density Relation [3]

$$P = \rho R_s T \tag{7}$$

Ideal Gas Law

$$Ma = \frac{v}{c} \tag{8}$$

Mach Velocity [3]

$$c = \sqrt{\gamma \frac{P}{\rho}} = \sqrt{\gamma R_s T} \tag{9}$$

Speed of Sound [3]

$$q = \frac{1}{2}\rho v^2 = \frac{\gamma}{2} P Ma^2 \tag{10}$$

Dynamic Pressure [3]

$$\epsilon = \left(\frac{\gamma + 1}{2}\right)^{\frac{\gamma+1}{2(\gamma-1)}} Ma_e \left(1 + \frac{\gamma - 1}{2}Ma_e^2\right)^{-\frac{\gamma+1}{2(\gamma-1)}} \tag{11}$$

Area Ratio [3]

$$P_c = \frac{\dot{m}\sqrt{T_t}}{A_t}\sqrt{\frac{R_s}{\gamma}}\left(\frac{\gamma+1}{2}\right)^{\frac{\gamma+1}{2(\gamma-1)}} \tag{12}$$

Chamber Pressure [4]

$$A_t = \frac{\dot{m}}{\rho_t v_t} \tag{13}$$

Throat Area [5]

$$0 = \frac{Ma_t}{Ma_e}\left(\frac{1+\left(\frac{\gamma-1}{2}Ma_e^2\right)}{1+\left(\frac{\gamma-1}{2}Ma_t^2\right)}\right)^{\frac{\gamma+1}{2(\gamma-1)}} - \epsilon \tag{14}$$

Isentropic Gas Relations [6]

References

[1] J. Hawley, "The stoichiometry of burning hydrocarbon fuels." https://jimhawley.ca/downloads/Diesel/The_stoichiometry_of_burning_hydrocarbon_fuels.pdf, 2020.

[2] O. Lanzi, "How do you calculate the heat capacity ratio for a multi-compound gas?." https://chemistry.stackexchange.com/a/166796, 2022.

[3] AMES Research Staff, "Equations, tables, and charts for compressible flow," *National Advisory Committee for Aeronautics*, 1953.

[4] G. R. Center, "Rocket thrust summary." https://www.grc.nasa.gov/WWW/k-12/airplane/rktthsum.html, 2021.

[5] B. Brady, "Cryo rocket." https://www.cryo-rocket.com, 2020.

[6] O. B. Robert D. Zucker, *Fundamentals of Gas Dynamics*. Wiley, 2019.